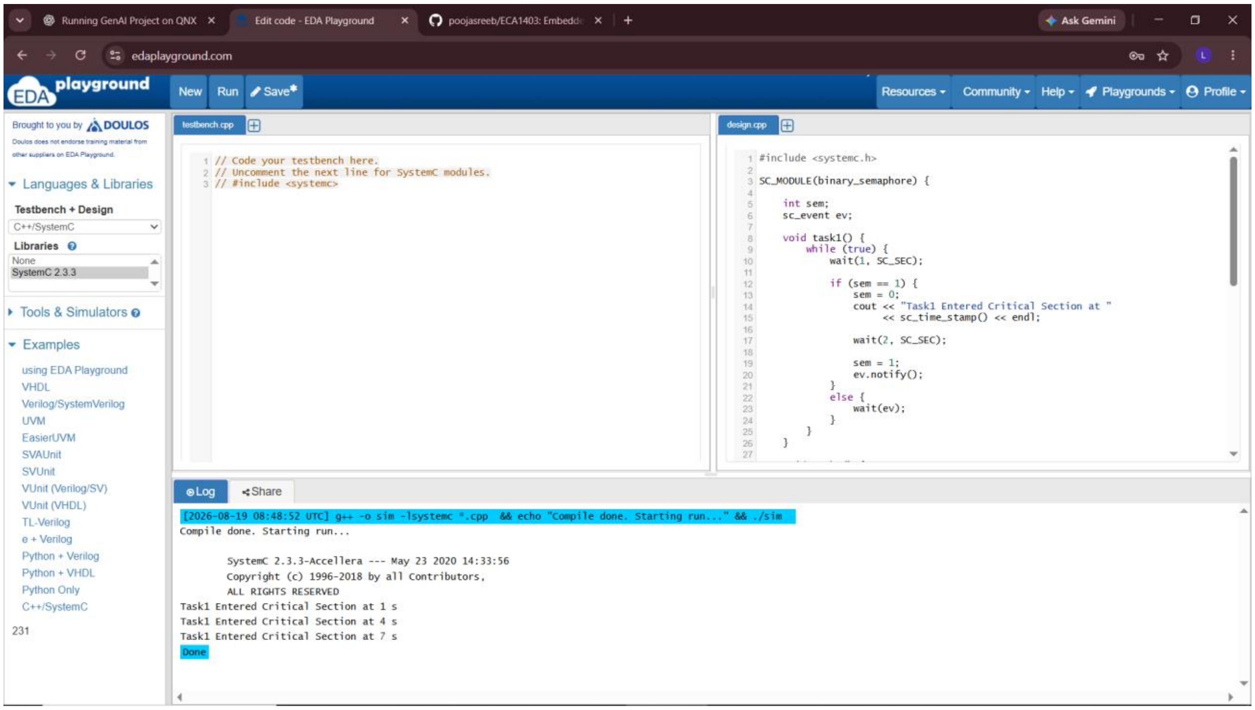

SOURCE CODE

```
#include <systemc.h>

SC_MODULE(binary_semaphore) {

   int sem;
   sc_event ev;

   void task1() {
      while (true) {
         wait(1, SC_SEC);

         if (sem == 1) {
            sem = 0;
            cout << "Task1 Entered Critical Section at "
                << sc_time_stamp() << endl;

            wait(2, SC_SEC);

            sem = 1;
            ev.notify();
         }
         else {
            wait(ev);
         }
      }
```

```
    }

    void task2() {
        while (true) {
            wait(2, SC_SEC);

            if (sem == 1) {
                sem = 0;
                cout << "Task2 Entered Critical Section at "
                     << sc_time_stamp() << endl;

                wait(1, SC_SEC);

                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }

    SC_CTOR(binary_semaphore) {
        sem = 1;
        SC_THREAD(task1);
        SC_THREAD(task2);
    }
};

int sc_main(int argc, char* argv[]) {
    binary_semaphore obj("Binary_Semaphore");
    sc_start(10, SC_SEC);

    return 0;
}
```